

HW1_22

Read data

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
## filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
## intersect, setdiff, setequal, union
```

```
setwd("/Users/chenzhiyi/Documents/碩班/碩一/財務計量經濟/HW1/")  
star <- read.csv("star5_small.csv")
```

```
#Check structure  
str(star)
```

```
## 'data.frame': 1200 obs. of 7 variables:  
## $ aide : int 0 0 1 1 0 1 0 0 0 1 ...  
## $ mathscore : int 528 454 463 449 478 536 444 528 478 459 ...  
## $ readscore : int 461 436 395 430 421 445 434 453 467 408 ...  
## $ regular : int 0 1 0 0 0 0 0 0 1 0 ...  
## $ small : int 1 0 0 0 1 0 1 1 0 0 ...  
## $ tchexper : int 13 8 8 3 15 11 12 6 11 8 ...  
## $ totalscore: int 989 890 858 879 899 981 878 981 945 867 ...
```

```
summary(star)
```

```
##      aide      mathscore      readscore      regular
## Min.   :0.0000   Min.   :339.0   Min.   :358.0   Min.   :0.0000
## 1st Qu.:0.0000   1st Qu.:454.0   1st Qu.:414.0   1st Qu.:0.0000
## Median :0.0000   Median :478.0   Median :433.0   Median :0.0000
## Mean   :0.3542   Mean   :485.9   Mean   :435.8   Mean   :0.3433
## 3rd Qu.:1.0000   3rd Qu.:513.0   3rd Qu.:451.0   3rd Qu.:1.0000
## Max.   :1.0000   Max.   :626.0   Max.   :627.0   Max.   :1.0000
##
##      small      tchexper      totalscore
## Min.   :0.0000   Min.   : 0.000   Min.   : 727.0
## 1st Qu.:0.0000   1st Qu.: 5.000   1st Qu.: 870.0
## Median :0.0000   Median : 9.000   Median : 915.0
## Mean   :0.3025   Mean   : 9.126   Mean   : 921.6
## 3rd Qu.:1.0000   3rd Qu.:13.000   3rd Qu.: 963.0
## Max.   :1.0000   Max.   :27.000   Max.   :1253.0
##
##      NA's      :6
```

控制組：普通班級 (Regular size, 22–25 人)。

實驗組 A：小班制 (Small size, 13–17 人)。

實驗組 B：普通班級 + 全職助教 (Regular + Aide)。

(a)

篩選資料：僅保留小班(small)與普通班(regular)，排除有助教的班級

```
data_a <- subset(star, aide==0)
#執行回歸
model_a <- lm(totalscore ~ small, data = data_a)
summary(model_a)
```

```
##
## Call:
## lm(formula = totalscore ~ small, data = data_a)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -189.44  -50.44   -9.44   43.06   324.38
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   916.442     3.675  249.396  <2e-16 ***
## small         12.175     5.369   2.268   0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 74.59 on 773 degrees of freedom
## Multiple R-squared:  0.006608,    Adjusted R-squared:  0.005323
## F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

$$\widehat{TOTALSCORE} = 916.442 + 12.175 \times SMALL$$

1. $\beta_1 = 916.442$ ，代表普通班級 (Regular Class) 學生的平均總分（當small = 0 時，預期得分約為 916.44

分。)

2. $\beta_2=12.175$ ，在其他條件相同的情況下，小班制學生的平均總分比普通班學生高出約 12.18 分。

3. P-value = 0.0236 < 0.05，代表顯著，拒絕「小班制對成績沒有影響」的虛無假設 ($H_0: \beta_2 = 0$)。

(b)

Repeat part (a) using dependent variables READSCORE and MATHSCORE. Do you observe any differences?

```
model_read <- lm(readscore ~ small, data = data_a)
model_math <- lm(mathscore ~ small, data = data_a)

summary(model_read)
```

```
##
## Call:
## lm(formula = readscore ~ small, data = data_a)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -74.665 -21.590  -2.665   16.410  194.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   432.665      1.488  290.760 < 2e-16 ***
## small          6.924       2.174   3.185  0.00151 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30.2 on 773 degrees of freedom
## Multiple R-squared:  0.01295,    Adjusted R-squared:  0.01167
## F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
summary(model_math)
```

```
##
## Call:
## lm(formula = mathscore ~ small, data = data_a)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -144.777 -35.028  -5.028   29.223  142.223
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   483.777      2.449  197.545 <2e-16 ***
## small          5.251       3.578   1.467  0.143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.71 on 773 degrees of freedom
## Multiple R-squared:  0.002778,    Adjusted R-squared:  0.001488
## F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

$$\widehat{READSCORE} = 432.665 + 6.924 \times SMALL$$

1. $\beta_1 = 432.665$ ，普通班學生的平均閱讀分數約為 432.67 分。
2. $\beta_2 = 6.924$ ，在小班制就讀的學生，其閱讀分數平均比普通班學生高出約 6.92 分。
3. $P\text{-value} = 0.00151 < 0.01$ ，小班制對閱讀分數的正面影響在 1% 的顯著水準下顯著。

$$\widehat{MATHSCORE} = 483.777 + 5.251 \times SMALL$$

1. $\beta_1 = 483.777$ ，普通班學生的平均數學分數約為 483.78 分。
2. $\beta_2 = 5.251$ ，在小班制中的學生，其數學分數平均比普通班高出 5.25 分。
3. $P\text{-value} = 0.143 < 0.05$ ，係數不顯著，無法拒絕「小班制對數學成績沒有影響」的虛無假設。

(c)

助教 (AIDE) 對總分的影響

```
# 篩選資料：僅保留普通班(regular)與有助教的普通班(aide)
data_c <- subset(star, regular == 1 | aide == 1)

# 執行回歸
model_c <- lm(totalscore ~ aide, data = data_c)
summary(model_c)
```

```
##
## Call:
## lm(formula = totalscore ~ aide, data = data_c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -191.748  -50.442   -5.442   40.558  312.558
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   916.442     3.559  257.529  <2e-16 ***
## aide           4.306     4.994   0.862    0.389
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 72.23 on 835 degrees of freedom
## Multiple R-squared:  0.0008898, Adjusted R-squared: -0.0003068
## F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888
```

$$\widehat{TOTALSCORE} = 916.442 + 4.306 \times SMALL$$

1. $\beta_1 = 916.442$ ，無助教班學生的平均總分約為 916.442 分。
2. $\beta_2 = 4.306$ ，在有助教班就讀的學生，其總分平均比無助教班學生高出約 4.306 分。
3. $P\text{-value} = 0.389 > 0.05$ ，不顯著，配置助教對學生成績幾乎沒有顯著性的幫助。

(d)

助教對閱讀與數學的分別影響

```
model_read_aide <- lm(readscore ~ aide, data = data_c)
model_math_aide <- lm(mathscore ~ aide, data = data_c)

summary(model_read_aide)
```

```
##
## Call:
## lm(formula = readscore ~ aide, data = data_c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -74.665 -21.536  -2.536   15.464  194.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   432.665      1.460  296.341  <2e-16 ***
## aide           2.871       2.049   1.401    0.161
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 29.64 on 835 degrees of freedom
## Multiple R-squared:  0.002347,    Adjusted R-squared:  0.001152
## F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615
```

```
summary(model_math_aide)
```

```
##
## Call:
## lm(formula = mathscore ~ aide, data = data_c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -146.212 -31.212  -5.777   29.223  142.223
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   483.777      2.355  205.464  <2e-16 ***
## aide           1.435       3.304   0.434    0.664
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 47.79 on 835 degrees of freedom
## Multiple R-squared:  0.0002258,    Adjusted R-squared:  -0.0009715
## F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

$$\widehat{READ} = 432.665 + 2.871 \times AIDE$$

1. $\beta_1 = 432.665$ ，無助教班學生的平均閱讀分數約為 432.665 分。
2. $\beta_2 = 2.871$ ，在有助教班就讀的學生，其閱讀分數平均高出約 2.871 分。
3. $P\text{-value} = 0.161 > 0.05$ ，不顯著，配置助教對學生成績幾乎沒有顯著性的幫助。

$$\widehat{MATH} = 483.777 + 1.435 \times AIDE$$

1. $\beta_1 = 483.777$ ，無助教班學生的平均數學分數約為 483.78 分。
2. $\beta_2 = 1.435$ ，在有助教班的學生，其數學分數平均比普通班高出 1.435分。
3. $P - value = 0.664 > 0.05$ ，不顯著，配置助教對學生成績幾乎沒有顯著性的幫助。